

Polad Ibrahimov

AI / Machine Learning Engineer

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Profile

AI/ML Engineer with 2.5+ years of production experience at Azerbaijan's National AI Center, now a Computer Science M.Sc. student at the University of Bonn. Specialised in retrieval-augmented generation, semantic search and embedding models; co-author of a research paper on legal AI and core contributor to *eqanun.ai*, a deployed legal AI assistant. Seeking a working student position (up to 20 h/week) in AI, ML or Data Science.

Professional Experience

Middle AI Engineer

Jan 2025 – Mar 2026

National AI Center — Baku, Azerbaijan

- Core contributor to **eqanun.ai**, a production legal AI assistant for Azerbaijani law.
- Designed hybrid retrieval and vector storage pipelines with **FAISS**, **Elasticsearch** and Vespa-style systems, combining dense and lexical search with cross-encoder reranking.
- Built robust **OCR** pipelines for complex legal PDFs — multi-column layouts, scans and tables.
- Cut latency and infrastructure cost via batching, caching and parallelisation; tuned quality through hyperparameter search and ablation studies.
- Deployed models as containerised microservices with **Docker**, **Kubernetes** and CI/CD.

AI Engineer (Intern → Junior)

Jun 2023 – Jan 2025

National AI Center — Baku, Azerbaijan

- Implemented semantic search and document ranking for low-resource (Azerbaijani) text, plus prototypes for document parsing and structured data extraction.
- Developed reusable **Python** pipelines to preprocess, clean and augment large-scale text corpora for training and evaluation.

Research

Research & Development of AI-Powered Legal Search

Apr 2024 – Mar 2025

National AI Center — Baku / supervised by the Head of AI

- Trained domain-specific embedding and reranking models on transformer architectures for Azerbaijani legal text, targeting query understanding, ranking and low-resource adaptation.
- Results shipped into the E-qanun AI platform and documented in a technical research paper.

Education

M.Sc. Computer Science

Sep 2025 – Aug 2027 (expected)

University of Bonn — Bonn, Germany

- Admitted through a paper-based research contest on large language models.

B.Sc. Computer Science — GPA 98/100

Sep 2021 – Jun 2025

French-Azerbaijani University (UFAZ), Baku — dual degree with the University of Strasbourg (France) and Azerbaijan State Oil and Industry University

Skills, Certifications & Languages

Programming: Python (advanced), Java (advanced), SQL, C (intermediate)

ML & AI: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, LLMs, RAG, embeddings, reranking, semantic search, OCR, reinforcement learning

Retrieval & Data: FAISS, Elasticsearch, Vespa, Pandas, NumPy, distributed data processing

Backend & MLOps: FastAPI, Django, REST APIs, Docker, Kubernetes, Git, CI/CD

Certifications: IELTS Academic 7.0 (2024); Neural Networks & Deep Learning — Andrew Ng, Coursera (2022)

Languages: Azerbaijani & Turkish (native), English (C1), Russian & French (B1), German (A1, learning)

Projects

Quantum-Inspired Reinforcement Learning for Finance

- Prototyped an RL agent combining quantum state representations with classical RL to improve exploration in financial decision-making; benchmarked on historical market data (GitHub).

Awards & Scholarships

- Erasmus Mundus Scholarship** — selected for the highly competitive international programme. **EU Grant Recipient** (Aug 2025). **Visegrad Scholarship** — declined in favour of studies in Germany.
- 1st Place, SOCAR Hackathon (2023)**; **2nd Place, UFAZ Hackathon (2022)**; **Technest Scholarship (2023)**; **Winner, Lutfi Zadeh Logic Olympiad (2017)**.